



Team Project 1 - CAP

Html Fundamental

What is HTML?

- HTML stands for HyperText Markup Language.
- It is the standard language used to create and structure content on the web.
- HTML provides the structure, while CSS provides style and JavaScript provides interactivity.
- Every web page you visit is built with HTML.

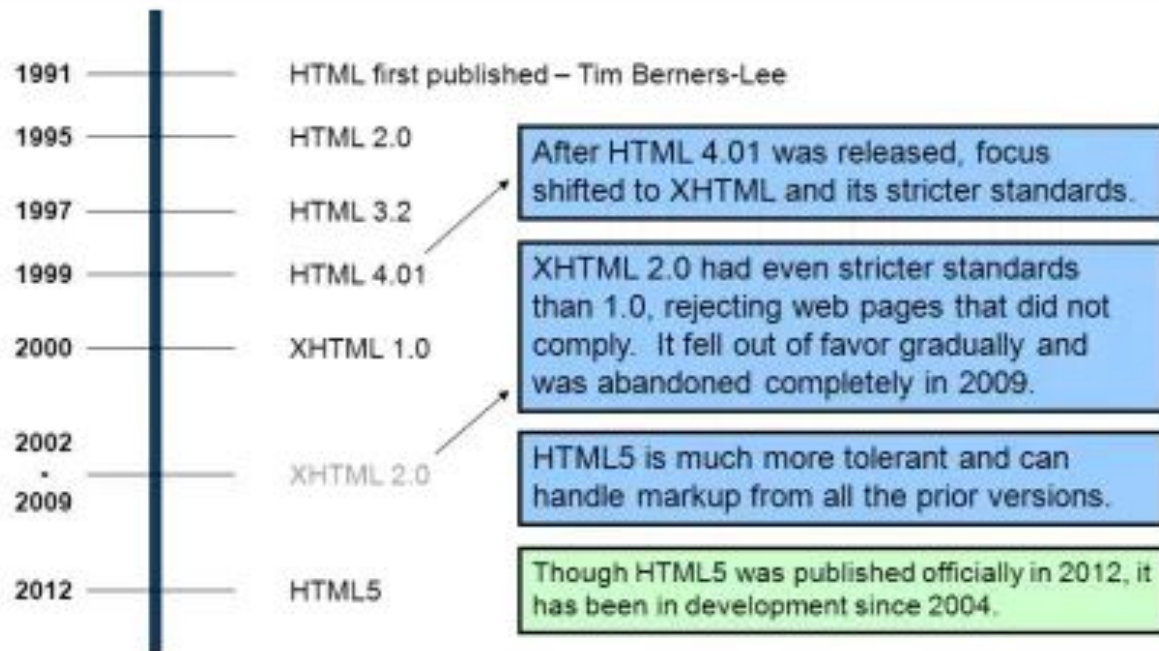
Example:

```
<!DOCTYPE html>  
  <html>  
    <head><title>My Page</title></head>  
    <body>Hello</body>  
</html>
```

What is HTML?



History of HTML



Anatomy of an HTML Document

- <!DOCTYPE html> → Document type
- <html> → Root element
- <head> → Metadata, script, styles, title, links
- <body> → Visible content (text, images, links, forms).

Example:

```
<html>
  <head>
    <meta charset="UTF-8">
    <title>Document Structure</title>
  </head>
  <body>
    <h1>Content goes here.</h1>
    <p>more</p>
  </body>
</html>
```

The <head> and <body>

Head contains: (Metadata about the page)

- Page Title
- Meta tags (charset, description, author ...etc)
- CSS & JavaScript links

Body contains:

- Text, images, forms (What user see)

Example:

```
<head>
  <title>My Website</title>
  <link rel="stylesheet" href="style.css">
  <script src="script.js"></script>
</head>
<body>
  <h1>Welcome!</h1> <p>This is the visible content.</p>
</body>
```

HTML Elements & Tags

- Elements = opening tag + content + closing tag
- Attributes add extra info

Example:

```
<p>This is a paragraph.</p>
```

```

```

```
<a href="https://example.com" target="_blank">
```

Visit Website

```
</a>
```

HTML Text Tags - Formatting

Tag	Description
	Defines bold text
	Defines emphasized text
<i>	Defines a part of text in an alternate voice or mood
<small>	Defines smaller text
	Defines important text
<sub>	Defines subscripted text
<sup>	Defines superscripted text
<ins>	Defines inserted text
	Defines deleted text
<mark>	Defines marked/highlighted text

HTML Comments

- Comments are **not displayed** in the browser.
- Used to **explain code** or leave notes for developers.
- Helpful for **documentation, debugging, or reminders**.
- Comments can hide code temporarily.

Example Code:

```
<!-- This is a single-line comment -->
```

```
<p>This is visible text.</p>
```

```
<!--  
This is a multi-line comment  
It can span multiple lines  
-->
```

```
<!-- <p>This paragraph is hidden because it is inside a  
comment</p> -->
```


Semantic HTML

- Semantic tags give meaning
- Improve SEO (search engines understand content better).
- Accessibility (screen readers).
- Code readability for Human and LLMs

Tags: <header>, <footer>, <article>, <section>, <nav>, <aside>.

Example:

```
<header>
  <h1>My Blog</h1>
  <nav>
    <a href="#">Home</a> | <a href="#">Articles</a>
  </nav>
</header>
<article>
  <h2>First Post</h2>
  <p>This is the blog post content.</p>
</article>
<footer>© 2025 My Blog</footer>
```

Text Content & Formatting

- Headings: <h1>.. <h6>
- Paragraphs: <p>
- Inline formatting: , ,

Example:

```
<h1>Main</h1>
```

```
<p><strong>Bold</strong>
```

```
<em>Italic</em></p>
```

Lists

- Ordered List: `<ol>` → numbered.
- Unordered List: `<ul>` → bulleted.
- List items: `<li>` inside lists.

Example:

```
<ul>  
  <li>Bread</li>  
  <li>Milk</li>  
</ul>
```

Hyperlinks <a>

- Connect pages together or external web url
- Attributes: href, target='_blank' or '_self' ...
- Can link to anchors with element id

Example:

```
<a href='https://example.com'>Link</a>
```

```
<a href='about.html'>Link</a>
```

```
<a href='#id_of_elm'>Link</a>
```

Images & Media

- : requires src & alt
- <audio>: controls attr
- <video>: controls attr

Example:

```

```

```
<audio controls>
```

```
  <source src="song.mp3" type="audio/mpeg">
```

```
</audio>
```

```
<video width="400" controls>
```

```
  <source src="movie.mp4" type="video/mp4">
```

```
</video>
```

Tables

- Use for structured data (not layout)
- Tags: <table>, <tr>, <th>, <td>

Example:

```
<table border="1">  
  <tr>  
    <th>Name</th><th>Age</th><th>Grade</th>  
  </tr>  
  <tr>  
    <td>Ali</td><td>20</td><td>A</td>  
  </tr>  
  <tr>  
    <td>Lina</td><td>21</td><td>B</td>  
  </tr>  
</table>
```

Forms & Inputs

- Forms collect user input
- Attributes: action (URL to send data), method (post, get)
- Inputs: text, email, checkbox, radio ...etc

Example:

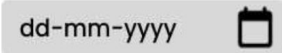
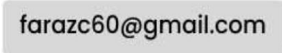
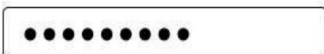
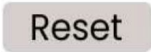
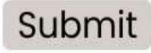
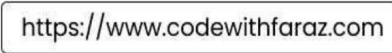
```
<form name="myForm" action="_self" method="post">  
  <input type='text'>  
  <input type='submit'>  
</form>
```

Forms & Inputs

Note that to submit the user input you need to add a button inside the form.

- See this link to learn about form elements:
https://www.w3schools.com/html/html_form_elements.asp
- See this link to learn about input types:
https://www.w3schools.com/html/html_form_input_types.asp

Forms & Inputs

<code><input type="button"></code>	
<code><input type="checkbox"></code>	
<code><input type="color"></code>	
<code><input type="date"></code>	
<code><input type="email"></code>	
<code><input type="file"></code>	
<code><input type="hidden"></code>	
<code><input type="image"></code>	
<code><input type="number"></code>	
<code><input type="password"></code>	
<code><input type="radio"></code>	
<code><input type="range"></code>	
<code><input type="reset"></code>	
<code><input type="submit"></code>	
<code><input type="text"></code>	
<code><input type="url"></code>	

HTML Tags – Buttons

- The `<button>` tag defines a clickable button inside the HTML document.
- `<button>` has an opening and closing tags (like `<a>`) which allows the developers to insert tags like: `<strong>`, `<img>`, `<i>`, .. Inside the `<button>`
- `<button>` has an attribute called “type” which tells the browser about that button’s behavior.
- The “type” attribute in the `<button>` has three options:
 - “button”: which allows the button to act like a normal button.
 - “submit”: used when adding a button inside `<form>` to send the data.
 - “reset”: resets all form values.

Labels

- The tag is used when you have a form or input elements in which, the label is associated with an input element via the “for” attribute that the input id.
- The element does not render as anything special for the user. However, it provides a usability improvement for mouse users, because if the user clicks on the text within the element, it toggles the control associated with it.
- The “for” attribute of the tag should be equal to the id attribute of the related element to bind them together. (we will discussed it in details in the Forms section).

Example:

```
<label>
```

```
    Username: <input type="text" name="username">
```

```
</label>
```

Accessibility

- Alt text for images
- Semantic tags help screen readers
- ARIA attributes

Example:

```
<button aria-label="Close">X</button>
```

```

```

Block-Level vs Inline HTML Elements

Block Elements

- Takes up the full width available, and has a line break before and after it.

Ex:

<p>, <div>, <h1>.... <h6>, , , , <table>

Inline Elements- Takes up only as much width as necessary, and does not force line breaks after it.

Ex:

<a>, ,

Video:

<https://www.youtube.com/watch?v=M4n-WSkehml>

HTML Best Practices

- Use semantic tags
- Close all elements
- Indentation & readability
- Separate HTML/CSS/JS
- Validate HTML at validator.w3.org

Hands-On Example

Exercise: Build an 'About Me' page

- Add heading
- Add paragraph
- Add image
- Add list of hobbies
- Add link to social media

Summary

- HTML = structure of webpage
- Semantic tags improve SEO & accessibility
- Forms, tables, media enrich content
- Follow best practices
- HTML + CSS + JS = foundation

Html Exercise – Build Your Own About Me Page

- Create a personal **About Me** webpage using **HTML only** (no CSS or JS required).
- Use **semantic HTML5 elements**: `<header>`, `<nav>`, `<section>`, `<article>`, `<footer>`.

Include the following sections:

1. **Header** with your name and a short tagline.
 2. **Navigation** links to each section.
 3. **About:** a short paragraph about yourself.
 4. **Education:** list your degrees or courses.
 5. **Experience:** jobs, projects, or volunteering.
 6. **Hobbies/Interests:** unordered list.
 7. **Table:** e.g., weekly schedule, favorite books, or skills with levels.
- Contact Info:** email, phone, or social media (use `<address>`).

- Use at least **one image** with an `alt` attribute.
- Ensure all tags are **properly nested, closed, and semantic**.
Validate your HTML at validator.w3.org.

Example Prompt (Doctor Hasan Ali):

Build an “About Me” page for **Dr. Hasan Ali**, including a biography, education list, clinic hours in a table, and contact details.

Dr. Hasan Ali, MD

Board Certified Internal Medicine Physician

- About
- Education & Certifications
- Clinical Interests
- Services
- Clinic Hours
- Publications
- Patient Feedback
- Contact & Location

About Dr. Ali

Dr. Hasan Ali, MD, is an internal medicine physician with over 12 years of experience in diagnosing and managing adult medical conditions, including hypertension, diabetes, thyroid disorders, and preventive care. He practices evidence-based medicine and emphasizes clear communication and shared decision-making.



Dr. Ali at his clinic in Nafdis.

Clinical Interests

- Cardiovascular health (hypertension, diabetes, dyslipidemia)
- Preventive medicine and screening
- Endocrine and metabolic disorders
- Chronic disease management

Education & Certifications

- MD (Doctor of Medicine), Aga Saeed National University Faculty of Medicine – 2011
- Residency: Internal Medicine, Nafdis Medical Center – 2012 – 2015
- Board Certification, Internal Medicine – 2016 (current)
- Certificate: Advanced Cardiac Life Support (ACLS) – May 2020

Clinical Experience

Internal Medicine Consultant — Nafdis Medical Center

2016-Present

- Department and supervisor management of adult medical conditions
- Chronic disease clinics for diabetes and hypertension
- Collaboration with radiology, endocrinology, and nutrition teams

Internal Medicine Specialist — Rafidra Governmental Hospital

2015-2019

- Acute care, admission, and ward rounds
- Medical education for junior doctors and interns
- Development of standardized care pathways for common conditions

Services

- Comprehensive adult health checkups and annual screenings
- Diabetes care (diagnosis, medication titration, lifestyle plans)
- Hypertension management and cardiovascular risk reduction
- Thyroid disorder evaluation and long-term follow-up
- Vaccination counseling and travel medicine advice

Clinic Hours & Appointment Types

Please arrive 10 minutes early for registration. Bring your medication list and any recent lab results.

Weekly Schedule (Nafdis Clinic)			
Day	Morning	Afternoon	Appointment Type
Sunday	09:00 - 12:00	13:00 - 16:00	New & Follow-up. Parking lots available.
Monday	09:00 - 12:00	13:00 - 16:00	Patient only. Patients: clinic hours.
Tuesday	09:00 - 12:00	—	New patients. Allow extra time for history.
Wednesday	09:00 - 12:00	13:00 - 17:00	Patient only. Hypertension clinic hours.
Thursday	09:00 - 12:00	—	Medical. Vaccinations by appointment.
Friday & Saturday	Closed (urgent cases via hospital on-call)		

Selected Publications

1. Ali H, et al. "Integrated Care Pathways for Type 2 Diabetes in Resource-Constrained Settings." *Journal of Primary Care*, 2023.
2. Ali H. "Hypertension Follow-up: Interventions in Community Clinics." *Middle East Journal of Medicine*, 2022.
3. Ali H, Smith R. "Physical Therapy Management for General Practitioners." *Clinical Practice Review*, 2021.

Patient Feedback

"Dr. Ali explained my treatment options clearly and helped me get my blood pressure under control."

— M.S., patient since 2020

"Thorough, kind, and very organized. Highly recommend for diabetes management."

— A.R., patient since 2019

Contact & Location

Nafdis Medical Center, Internal Medicine Clinic

Al-Bal, Jeddah, Building C, 2nd Floor

Nafdis, Jeddah

Phone: +966 29 999 9999

Email: hasan.ali@nafdis.com

For referrals, please include recent labs and imaging reports. Secure fax available upon request.



Clinic entrance and signage

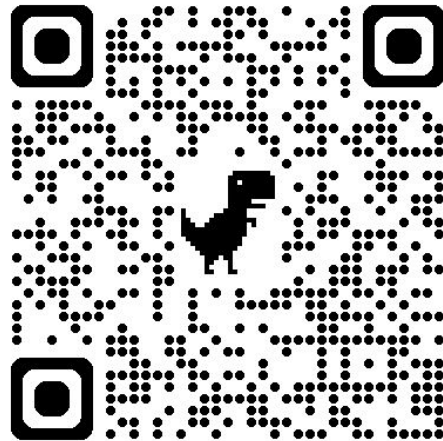
© 2023 Dr. Hasan Ali — Internal Medicine

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Exercise: Debugging HTML

The following HTML has a number of bugs in it. Use the code editor and the browser to find and resolve all the errors. Good luck!

<https://github.com/zeiadhabbab/Web-Fundamentals/blob/main/debug-html-exercise.html>



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